Zoology Legends

Zoo202

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Zoology Legends

Short & Long Questions...

1. Negative pressure mechanism

In most reptiles, a negative-pressure mechanism is responsible for lung ventilation. A posterior movement of the ribs and the body wall expands the body cavity, decreasing pressure in the lungs and drawing air into the lungs.

Air is expelled by elastic recoil of the lungs and forward movements of the ribs and body wall, which compress the lungs.

The ribs of turtles are a part of their shell; thus, movements of the body wall to which the ribs attach are impossible. Turtles exhale by contracting muscles that force the viscera upward, compressing the lungs. They inhale by

contracting muscles that increase the volume of the visceral cavity, creating negative pressure to draw air into the lungs.

2. Development and reproduction in cephalochordate

Reproduction:

• Cephalochordates are dioecious.

• Gonads bulge into the atrium from the lateral body wall.

• Gametes are shed into the atrium and leave the body through the atriopore.

Development:

• External fertilization leads to a bilaterally symmetrical larva.

• Larvae are free-swimming

• But they eventually settle to the substrate before metamorphosing into adults

3. Define spectacle

• In snakes and some lizards, the upper and lower eyelids fuse in the embryo to form a protective window of clear skin, called the spectacle. During ecdysis, the outer layers of the spectacle become clouded and impair the vision of snakes.

4. Define Metamorphosis

• Metamorphosis is a series of abrupt structural, physiological, and behavioral changes. Crowding and food availability, influence the time required for metamorphosis.

• Metamorphosis's under the control of neurosecretions of the hypothalamus

5. What is Ventriculus and proventriculus

The stomach of birds is modified into two regions.

Ventriculus

The ventriculus, or muscular stomach (also known as the "gizzard"), contains two opposing sets of muscles used for grinding food. It has muscular walls to abrade and crush seeds or other hard materials

Proventriculus

The proventriculus, or glandular stomach, is continuous with the esophagus at the level of the base of the heart and contains digestive (pepsinogen-secreting) and mucous glands.

The proventriculus secretes gastric juices that initiate digestion

6. Non shivering thermogenesis

• **Non-shivering thermogenesis** involves heat production by general cellular metabolism and the metabolism of special fat deposits called brown fat

• **Shivering thermogenesis** is muscular activity that generates large amounts of heat but little movement

7. Types of fight of Bird

Different birds, or the same bird at different times, use different kinds of flight

• During **gliding flight**, the wing is stationary, and a bird loses altitude. **Example**: Waterfowl

• **Flapping flight** generates the power for flight and is the **most common** type of flying. Many variations in wing shape and flapping patterns

result in species-specific speed and maneuverability

• **Soaring flight** allows some birds to remain airborne with little energy expenditure. During soaring, wings are essentially stationary, and the bird utilizes updrafts and air currents to gain altitude.

Example: Hawks, vultures, and other soaring birds.

• **Hummingbirds** perform **Hovering flight.** They hover in still air by fanning their wings back and forth **(50 to 80 beats per second)** to remain suspended in front of a flower or feeding station.

8. Altricial & Precocial young's

Altricial young's

Those birds that are entirely dependent on their parents are said to be **altricial** and they are often naked at hatching

• Altricial young must be brooded constantly at first because endothermy is not developed. They grow rapidly, and when they leave the nest, they are nearly as large as their parents.

For example, American robins weigh **4 to 6 g** at hatching and leave the nest **13 days** later weighing **57 g.**

Precocial young's

• Precocial young are alert and lively at hatching They are usually covered with down and can walk, run, swim, and feed themselves—

although one parent is usually present to lead the young to food and shelter.

Example: Killdeer

9. Advantage of viviparity

Viviparity is advantageous because females are not necessarily tied to a single nest site, but can roam or migrate to find food or a proper climate

• Viviparity is accompanied by the evolution of a portion of the reproductive tract where the young are nourished and develop

• In viviparous mammals, the oviducts are modified into one or two uteri

10. Mechanism of Embryonic Diapaus

• Mating occurs approximately two weeks after the birth of the previous year's off-spring.

• Development is arrested for the three months during which the recently born young do most of their nursing. This mechanism is called embryonic diapause

• Thus, even though actual development takes about nine months, the female carries the embryo and fetus for a period of one year.

This embryonic diapause, which was described previously for sea lions, also occurs in some bats, bears, martens, and marsupials.

11. Preening

Birds maintain a clean plumage to rid the feathers and skin of parasites. **Preening**, which is done by rubbing the bill over the feathers, keeps the feathers smooth, clean, and in place. **Hamuli** that become dislodged can be rehooked by running a feather through the bill.

12. Homodont & hetrodont

• In **reptiles**, the teeth are uniformly conical, a condition referred to as **homodont**

• In **mammals**, the teeth are often specialized for different functions, a condition called **heterodont.**

13. Mammary glands of Mammals

• Mammary glands are **functional in female** mammals and are present, but nonfunctional, in males.

• The milk that mammary glands secrete contains water, carbohydrates (especially the sugar lactose), fat, protein, minerals, and antibodies.

• Mammary glands are probably derived evolutionarily from **apocrine glands** and usually contain substantial fatty deposits

• **Monotremes** have mammary glands that lack nipples. The glands discharge milk into depressions on the belly, where the young lap it up

• In other mammals, mammary glands open via **nipples or teats**, and the young suckle for their nourishment.

14. Reproduction in tunicates or Urochordates

• Urochordates are **monoecious**. Gonads are located near the loop of the intestine. Genital ducts open near the atrial siphon.

• Gametes may be shed through the atrial siphon **for external fertilization** or eggs may be retained in the atrium for fertilization and early development.

• Although **self-fertilization occurs** in some species, **cross fertilization** is the rule.

15. Order of reptiles

They have four orders named as;

1. Testudines (Turtules)
2. Rhynchocephalia (Tuatara)
3. Squamata (The Lizards)
4. Crocodilia (Aligators, Crocodiles, Gavials Caimans)

16. Characteristics of mammals

• Any member of the group of vertebrate animals in which the young are nourished with milk from special mammary glands of the mother

• **Hair** is a typical mammalian feature, although in many **whales** it has disappeared except in the fetal stage

- A chain of **three tiny bones** transmits sound waves across the **middle ear**
- A muscular **diaphragm** separates the heart and the lungs from the abdominal cavity.
- Mature red blood cells (**erythrocytes**) in all mammals lack a **nucleus**; all other vertebrates have nucleated red blood cells
- The skin of mammals is thick and **protective** and has an insulating covering of hair.
- Adaptations of teeth and the digestive tract allow mammals to exploit a wide variety of food resources.
- **Metanephric** kidneys permit mammals to excrete **urea** without excessive water loss.
- Complex behavior patterns enhance mammalian survival

• Most mammals are **viviparous** and have reproductive cycles that help ensure **internal fertilization** and successful development.

17. Countercurrent exchange system

• Countercurrent heat-exchange systems may help regulate heat loss from exposed areas

• Arteries passing peripherally through the core of an appendage are surrounded by veins that carry blood back toward the body

• When blood returns to the body through these veins, heat transfers from arterial blood to venous blood and returns to the body rather than being lost to the environment

• When excess heat is produced, blood is shunted away from the countercurrent veins toward peripheral vessels, and excess heat is radiated to the environment

• **For Example**, during winter, the lower part of a **reindeer's leg** may be at 10°C, while body temperature is about 40°C.

18. Convergent evolution

• The evolution of similar structures in different lineages is called **convergent evolution**

• The mammalian heart evolved in the synapsid reptilian lineage, whereas the avian heart evolved in the **archosaur lineage**.

19. Mechanisms of inspiration and expiration in Mammals

Unlike reptiles and birds, however, mammals possess a **muscular diaphragm** that separates the thoracic and abdominal cavities

• **Inspiration** results from contraction of the diaphragm and expansion of the rib cage, both

of which decrease the intrathoracic pressure and allow air to enter the lungs

• **Expiration** is normally by elastic recoil of the lungs and relaxation of inspiratory muscles, which decreases the volume of the thoracic cavity

The contraction of other thoracic and abdominal muscles can produce forceful exhalation.

20. Hibernation with example? (3)

"**Hibernation** is a period of winter inactivity in which the hypothalamus of the brain slows the metabolic, heart, and respiratory rates"

• True hibernators include the **monotremes** (echidna and duck-billed platypus) and many members of the **Insectivora** (e.g., moles and shrews), **Rodentia** (e.g., chipmunks and woodchucks), and **Chiroptera (bats)**

• In preparation for hibernation, mammals usually accumulate large quantities of **body fat**.

• The **respiratory rate** of a hibernating ground **squirrel** falls from **100 to 200** breaths per minute to about **four breaths** per minute

• The heart rate falls from 200 to 300 beats per minute to about 20 beats per minute

• During hibernation, a mammal may lose a **third to half** of its body weight

21. Define Chamera?

• A frequently studied example, **Chimaera**, has a large head with a small mouth surrounded by large lips. A narrow, tapering tail has resulted in the common name **“ratfish”**

22. Arrector pili muscle? (5)

• A band of smooth muscle, called the **Arrector pili muscle**, runs between the hair follicle and the lower epidermis

• When the muscle contracts, the hairs stand upright, increasing the amount of air trapped in the pelage and improving its insulating properties.

• Arrector pili muscles are under the control of the autonomic nervous system, which also controls a mammal's "fight-or-flight" response

• In threatening situations, the hair (especially on the neck and tail) stands on end and may give the perception of increased size and strength.

23. Lateral line system

• Fishes has a lateral-line system.

• The lateral-line system consists of sensory pits in the epidermis. In these pits are receptors that are stimulated by water

• Lateral lines are used to detect either water currents, or a predator

• Fishes may also detect low-frequency sounds with these receptors

24. Characteristics of cartilaginous fishes

Class Chondrichthyes: Cartilaginous fishes

- Endoskeleton of cartilage
- Paired fins, movable jaws, gill slits
- Rough sandpaper – like skin
    - Placoid scales

- Pointed tip that is directed backward
- Same composition as teeth
- **Example**: Sharks, rays, skates, and ratfishes.

25. Sudoriferous gland ki two types

• Most mammals also possess sudoriferous (**sweat**) glands

• **Small sudoriferous** glands (eccrine glands) release watery secretions used in evaporative cooling

• **Larger sudoriferous** glands (apocrine glands) secrete a mixture of salt, urea, and water, which microorganisms on the skin convert to odorous products.

26. Difference between alligator and crocodile

Comparison Table Crocodile vs Alligator

Characters	Crocodile	Alligator
Distribution	Asia, Africa, Central America, Malaya, Indonesia and North Australia.	Mostly found in North America and China.
Habitats	Crocodile live in freshwater, They can also live in salt water.	Alligator live in fresh water. Lakes, swamps and streams have fresh water.
Habits	More aggressive, dangerous to man.	Less aggressive, attack when provoked.
Four mandibular tooth	First into a maxillary groove and visible externally.	Fits into a pit and not visible externally.
Teeth	Unequal.	Very unequal.
Body length	8 metres long.	3 metres long.
Color	Olive green with black spot or bands.	Steel grey.
Snout	Moderately long and pointed.	Short and broad.
	Long pointed jaw - V shaped.	Wide round jaw - U shaped.
www.majordifferences.com	 V shaped snout designed for snatching prey.	 U shaped snout designed for more leverage for crushing.

27. Define tunic

The body wall of most tunicates (L. tunicatus, to wear a tunic or gown) is a connective-tissue-like covering, called the tunic, which appears gel-like but is often quite tough.

• Secreted by the **epidermis**, it is composed of proteins, various salts, and cellulose.

28. Feeding of Enteropneusta

• Cilia and mucus assist acorn worms in feeding.
• Detritus and other particles adhere to the mucus-covered proboscis. Tracts of cilia transport food and mucus posteriorly and ventrally.

• Ciliary tracts converge near the mouth and form a mucoid string that enters the mouth.

• Acorn worms may reject some substances trapped in the mucoid string by pulling the proboscis against the collar.

• Ciliary tracts of the collar and trunk transport rejected material and discard it posteriorly.

29. Write the partition of atrium in Urodeles and Anurans

Atrium is **partially** divided in urodeles and **completely** divided in anurans.

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Remember me in you Precious Prayer

Jazak Allah